

Adriana Garcia

☎ +1 (703) 855-1518 ✉ adrianagarbol@gmail.com 🔗 linkedin.com/in/adrianagarbol

Education

- **Virginia Polytechnic Institute and State University** Aug 2025 – Fall 2027
Bachelor of Science in Electrical Engineering
- **Northern Virginia Community College** 2023 – May 2025
Associate of Science in Engineering GPA: 3.75

Technical Skills

Tools: KiCad, LTspice, MATLAB, Arduino, Tableau, Microsoft Office

Programming: C/C++, Python

Technical Areas: PCB Design, Embedded Systems, Analog Circuits, Control Systems, Sensor Integration

Languages: English, Spanish

Projects

- **Home Audio System (Analog Equalizer and Amplification)**
 - Designed and build a 3 bandpass filter equalizer and a Class D amplifier.
 - Analyzed circuit behavior using LTspice simulations and validated performance through physical testing.
 - Iteratively debugged and tuned circuit performance through multiple design revisions.
- **Smart Light Sensor PCB (KiCad)**
 - Designed a PCB in KiCad for alighting system using an LDR sensor and hysteresis comparator.
 - Implemented transistor-based switching to drive LEDs based on ambient light conditions.
 - Verified circuit behavior through simulation and real-world testing.
- **Bridge Automation System (Arduino)**
 - Designed and programmed an automated bridge model using an Arduino microcontroller and DC motors.
 - Implemented PWM-based motor control for speed regulation and digital logic for directional movement.
 - Collaborated in a team to integrate mechanical and electrical subsystems into a functional prototype.

Work Experience

- **Systems Specialist I** Jan 2025 – Aug 2025
Albireo Energy, Sterling, VA
 - Supported Building Automation Systems (BAS) in mission-critical data centers including Microsoft, CloudHQ, and Iron Mountain.
 - Diagnosed control wiring and communication issues between sensors, actuators, and controllers.
 - Interpreted electrical schematics and wiring diagrams to troubleshoot and optimize system performance.
 - Tested circuits using multimeters and tools to validate system functionality and safety.
- **Mathematics Tutor** Aug 2023 – Dec 2024
Northern Virginia Community College
 - Tutored students in Calculus I & II, Physics, and Precalculus.
 - Developed individualized strategies to improve student understanding and problem-solving skills.

Leadership and Activities

- **Program Scholar** 2023 – Present
NVOLVE
 - Participate in a selective STEM program focused on leadership and technical development.
- **Vice President of Operations**
NOVA Aviators Club
 - Organized technical workshops and collaborated with aviation professionals to enhance student engagement.